Dividing with Unit Fractions

Share a piece, or count how many pieces fit.

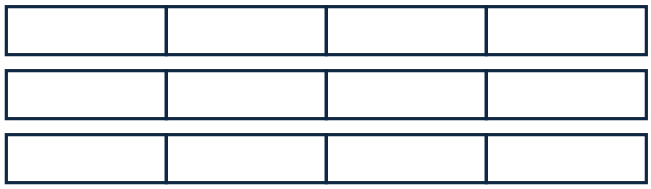
1/3 ÷ 2: share one third between 2 people



Cut every third into 2.
Now the whole has 6 pieces.
Each person gets 1 of them.

$$\frac{1}{3} \div 2 = \frac{1}{6}$$

3 ÷ 1/4: how many fourths fit in 3 wholes?



Each whole has 4 fourths.
3 wholes have 3 × 4 = 12 fourths.

$$3 \div \frac{1}{4} = 12$$

Check with times

$$\frac{1}{6} \times 2 = \frac{2}{6} = \frac{1}{3}$$

$$12 \times \frac{1}{4} = \frac{12}{4} = 3$$

Your turn!

$$\frac{1}{2} \div 4 = \frac{\square}{\square}$$

Dividing with Unit Fractions: Practice

Share a piece: it gets smaller. Pieces in wholes: count them.

Part 1 Share a piece Every piece is cut. What part of the whole is one small piece?

1) 

Share the shaded part among 2.

$\frac{1}{4} \div 2 = \frac{\square}{\square}$

2) 

Share the shaded part among 3.

$\frac{1}{2} \div 3 = \frac{\square}{\square}$

3) 

Share the shaded part among 3.

$\frac{1}{3} \div 3 = \frac{\square}{\square}$

4) 

Share the shaded part among 2.

$\frac{1}{5} \div 2 = \frac{\square}{\square}$

5) 

Share the shaded part among 5.

$\frac{1}{2} \div 5 = \frac{\square}{\square}$

Part 2 Share a piece No help this time.

6) $\frac{1}{4} \div 3 = \frac{\square}{\square}$

7) $\frac{1}{2} \div 6 = \frac{\square}{\square}$

8) $\frac{1}{5} \div 4 = \frac{\square}{\square}$

9) $\frac{1}{3} \div 5 = \frac{\square}{\square}$

10) $\frac{1}{6} \div 2 = \frac{\square}{\square}$

11) $\frac{1}{8} \div 2 = \frac{\square}{\square}$

Dividing with Unit Fractions: Practice (continued)

Part 2 Share a piece (continued)

12) $\frac{1}{2} \div 7 = \frac{\square}{\square}$

13) $\frac{1}{4} \div 5 = \frac{\square}{\square}$

14) $\frac{1}{3} \div 6 = \frac{\square}{\square}$

15) $\frac{1}{10} \div 3 = \frac{\square}{\square}$

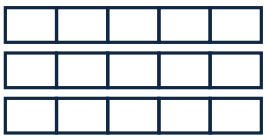
16) $\frac{1}{5} \div 5 = \frac{\square}{\square}$

17) $\frac{1}{6} \div 4 = \frac{\square}{\square}$

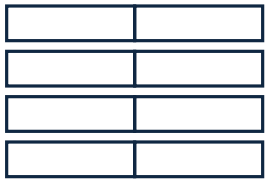
Part 3 How many pieces fit? Count all the pieces.

18) 

$2 \div \frac{1}{3} = \square$

19) 

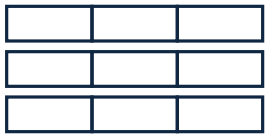
$3 \div \frac{1}{5} = \square$

20) 

$4 \div \frac{1}{2} = \square$

21) 

$2 \div \frac{1}{6} = \square$

22) 

$3 \div \frac{1}{3} = \square$

Dividing with Unit Fractions: Practice (continued)

Part 4 How many pieces fit? No help this time.

23) $5 \div \frac{1}{2} = \square$

24) $3 \div \frac{1}{6} = \square$

25) $6 \div \frac{1}{4} = \square$

26) $2 \div \frac{1}{8} = \square$

27) $7 \div \frac{1}{3} = \square$

28) $4 \div \frac{1}{5} = \square$

29) $8 \div \frac{1}{2} = \square$

30) $5 \div \frac{1}{6} = \square$

31) $9 \div \frac{1}{3} = \square$

32) $2 \div \frac{1}{10} = \square$

33) $6 \div \frac{1}{5} = \square$

34) $4 \div \frac{1}{8} = \square$

Dividing with Unit Fractions: Practice (continued)

Part 5 Check with times No help this time.

35) $\frac{1}{10} \times 5 = \frac{\square}{10}$

36) $\frac{1}{12} \times 4 = \frac{\square}{12}$

37) $\frac{1}{15} \times 3 = \frac{\square}{15}$

38) $\frac{1}{24} \times 6 = \frac{\square}{24}$

39) $\frac{1}{18} \times 3 = \frac{\square}{18}$

40) $\frac{1}{18} \times 9 = \frac{\square}{18}$

Part 6 Share a piece

41) $\frac{1}{7} \div 5 = \frac{\square}{\square}$

42) $\frac{1}{10} \div 4 = \frac{\square}{\square}$

43) $\frac{1}{4} \div 8 = \frac{\square}{\square}$

44) $\frac{1}{6} \div 8 = \frac{\square}{\square}$

45) $\frac{1}{9} \div 7 = \frac{\square}{\square}$

46) $\frac{1}{2} \div 3 = \frac{\square}{\square}$

Dividing with Unit Fractions: Practice (continued)

Part 6 Share a piece (continued)

47) $\frac{1}{3} \div 9 = \frac{\square}{\square}$

48) $\frac{1}{6} \div 9 = \frac{\square}{\square}$

49) $\frac{1}{8} \div 7 = \frac{\square}{\square}$

50) $\frac{1}{7} \div 6 = \frac{\square}{\square}$

51) $\frac{1}{7} \div 9 = \frac{\square}{\square}$

52) $\frac{1}{6} \div 6 = \frac{\square}{\square}$

53) $\frac{1}{9} \div 8 = \frac{\square}{\square}$

54) $\frac{1}{10} \div 8 = \frac{\square}{\square}$

55) $\frac{1}{4} \div 7 = \frac{\square}{\square}$

56) $\frac{1}{7} \div 3 = \frac{\square}{\square}$

57) $\frac{1}{8} \div 5 = \frac{\square}{\square}$

58) $\frac{1}{4} \div 6 = \frac{\square}{\square}$

Dividing with Unit Fractions: Practice (continued)

Part 6 Share a piece (continued)

59) $\frac{1}{9} \div 5 = \frac{\square}{\square}$

60) $\frac{1}{2} \div 5 = \frac{\square}{\square}$

61) $\frac{1}{9} \div 9 = \frac{\square}{\square}$

62) $\frac{1}{8} \div 6 = \frac{\square}{\square}$

63) $\frac{1}{7} \div 8 = \frac{\square}{\square}$

64) $\frac{1}{2} \div 9 = \frac{\square}{\square}$

Part 7 How many pieces fit?

65) $3 \div \frac{1}{7} = \square$

66) $6 \div \frac{1}{7} = \square$

67) $2 \div \frac{1}{9} = \square$

68) $8 \div \frac{1}{10} = \square$

69) $12 \div \frac{1}{7} = \square$

70) $7 \div \frac{1}{8} = \square$

Dividing with Unit Fractions: Practice (continued)

Part 7 How many pieces fit? (continued)

71) $10 \div \frac{1}{5} = \square$

72) $8 \div \frac{1}{8} = \square$

73) $9 \div \frac{1}{6} = \square$

74) $8 \div \frac{1}{9} = \square$

75) $9 \div \frac{1}{9} = \square$

76) $4 \div \frac{1}{4} = \square$

77) $7 \div \frac{1}{10} = \square$

78) $5 \div \frac{1}{8} = \square$

79) $7 \div \frac{1}{4} = \square$

80) $9 \div \frac{1}{4} = \square$

81) $12 \div \frac{1}{9} = \square$

82) $9 \div \frac{1}{2} = \square$

Dividing with Unit Fractions: Practice (continued)

Part 7 How many pieces fit? (continued)

83) $4 \div \frac{1}{7} = \square$

84) $12 \div \frac{1}{6} = \square$

85) $10 \div \frac{1}{3} = \square$

86) $2 \div \frac{1}{3} = \square$

87) $12 \div \frac{1}{10} = \square$

88) $10 \div \frac{1}{7} = \square$

Part 8 Remember? Problems from earlier packets.

89) $5 \div 7 = \frac{\square}{\square}$

90) $\frac{5}{6} \times 12 = \square$

91) $\frac{5}{7} - \frac{1}{3} = \frac{\square}{\square}$

92) $6 \times 0.35 = \square$

93) $2.4 + 3.75 = \square$

94) $828 \div 36 = \square$

Dividing with Unit Fractions: Practice (continued)

Part 9 Word problems Share it, or how many fit?

95) Half a pizza is shared equally by 3 friends. What part of the whole pizza does each get?

Number sentence _____

Answer ☐ of the pizza

Draw or show your work

96) How many 1/4 cup scoops are in 5 cups of rice?

Number sentence _____

Answer ☐ scoops

Draw or show your work

97) A 1/3 kg bag of nuts is split into 4 equal bags. How much is in each bag?

Number sentence _____

Answer ☐ kg

Draw or show your work

98) A 6 m rope is cut into pieces 1/2 m long. How many pieces are there?

Number sentence _____

Answer ☐ pieces

Draw or show your work

99) Challenge: 3 cups of flour, 1/3 cup per cake. Ava made 4 cakes. How many more can she make?

Number sentence _____

Answer ☐ cakes

Draw or show your work

Check-In: Dividing with Unit Fractions

Show what you know. No help this time. Score: _____ / 7

Part 1 Share a piece

1) 

$\frac{1}{2} \div 4 = \frac{\square}{\square}$

Share the shaded part among 4.

Part 2 Share a piece No help.

2) $\frac{1}{3} \div 7 = \frac{\square}{\square}$

3) $\frac{1}{4} \div 4 = \frac{\square}{\square}$

4) $\frac{1}{5} \div 3 = \frac{\square}{\square}$

Part 3 How many pieces fit?

5) $6 \div \frac{1}{2} = \square$

6) $3 \div \frac{1}{4} = \square$

7) $4 \div \frac{1}{6} = \square$